

Deterministic grids beat the Gaussian bad set for hyperplane tessellations

New partial result for the conjecture in arXiv:1111.4452

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Status. **New partial result; the original deterministic conjecture remains open.** The 2025 counterexample to the Gaussian hyperplane-tessellation conjecture is not a counterexample to Plan–Vershynin’s original, not-necessarily-Gaussian conjecture. In fact, the same lifted thin-cylinder set which needs order η^{-3} iid Gaussian hyperplanes has an explicit deterministic η -uniform tessellation using only order η^{-1} hyperplanes, and this deterministic order is sharp.

The two conjectures

For $K \subset \mathbb{S}^{N-1}$, let $m(K, \eta)$ be the least number of homogeneous hyperplanes for which the fraction separating any $x, y \in K$ approximates their normalized geodesic distance

$$d_{\mathbb{S}}(x, y) = \pi^{-1} \arccos \langle x, y \rangle$$

within additive error η . Plan and Vershynin [1] conjectured that

$$\sup_{\eta > 0} \eta^2 m(K, \eta) \asymp w_*(K)^2, \quad w_*(K) = \mathbb{E} \sup_{x \in K} |\langle g, x \rangle|.$$

Their upper conjecture permits arbitrary hyperplanes.

Dirksen and Strachan [2] isolate the later, stronger claim that $\eta^{-2} w_*(K)^2$ iid Gaussian hyperplanes always suffice. They disprove that claim by constructing sets S_η for which the Gaussian threshold is $\Theta(\eta^{-3} w_*(S_\eta)^2)$. Their introduction explicitly distinguishes this from the original not-necessarily-Gaussian question.

An explicit thin-cylinder theorem

For $\lambda > 0$, define the spherical lift

$$Q_\lambda(u) = \frac{(u, \lambda)}{\sqrt{\|u\|_2^2 + \lambda^2}}.$$

Theorem 1 (axis-plane grid). *Let $a, r, \lambda > 0$, $n \geq 1$, and*

$$S = Q_\lambda([-a, a] \oplus rB_2^n) \subset \mathbb{S}^{n+1}.$$

For every integer $m \geq 1$, there is an explicit arrangement A_m of m homogeneous hyperplanes such that

$$\sup_{x, y \in S} |d_{A_m}(x, y) - d_{\mathbb{S}}(x, y)| \leq \frac{2}{m} + \frac{2r}{\lambda}.$$

Consequently, if $2r/\lambda \leq \eta/2$, then $m(S, \eta) \leq \lceil 4/\eta \rceil$.

Proof. Let P be the two-dimensional plane spanned by the cylinder axis and the lifting coordinate. On the projective circle of homogeneous hyperplanes in P , choose m equally spaced normals. For any two points u, v on the great circle $\mathbb{S}^{n+1} \cap P$, a hyperplane separates them exactly when its projective zero lies in the interval between them. Counting an equally spaced grid in an arbitrary projective interval gives

$$|d_{A_m}(u, v) - d_{\mathbb{S}}(u, v)| \leq 2/m, \quad (1)$$

where the harmless constant 2 covers both endpoint conventions.

Write

$$x = Q_\lambda(t, z), \quad x_0 = Q_\lambda(t, 0).$$

Every selected normal lies in P , so

$$\text{sign}\langle h, x \rangle = \text{sign}\langle h, x_0 \rangle.$$

Thus the sign map ignores the orthogonal fiber *exactly*. Moreover, normalization on vectors of norm at least λ gives

$$\|x - x_0\|_2 \leq \frac{2\|z\|_2}{\lambda}.$$

For unit vectors, normalized geodesic distance is at most half their chord distance; hence $d_{\mathbb{S}}(x, x_0) \leq r/\lambda$.

For $y = Q_\lambda(s, w)$ and $y_0 = Q_\lambda(s, 0)$, the triangle inequality yields

$$|d_{\mathbb{S}}(x, y) - d_{\mathbb{S}}(x_0, y_0)| \leq d_{\mathbb{S}}(x, x_0) + d_{\mathbb{S}}(y, y_0) \leq 2r/\lambda.$$

Since $d_{A_m}(x, y) = d_{A_m}(x_0, y_0)$, combining this with (1) proves the result. $\square$

The proof gives a more general transfer principle: whenever a retraction $\rho : S \rightarrow \Gamma$ moves points by at most r_0 and the selected hyperplanes assign the same sign to x and $\rho(x)$, an ε -uniform tessellation of Γ becomes an $(\varepsilon + 2r_0)$ -uniform tessellation of S .

The 2025 Gaussian bad set is deterministically easy

We now unwind the parameters in the proof of Theorem 1.4 of [2]. Write η for its final spherical accuracy and set

$$q = \sqrt{4\pi C_0 \eta}, \quad \lambda = \frac{2C_0}{q}.$$

The constructed set has the form

$$S_\eta = Q_\lambda([-3q, 3q] \oplus r_\eta B_2^{n(\eta)}), \quad r_\eta = O\left(\frac{\eta}{\log(1/\eta)}\right).$$

Here the necessarily positive logarithmic denominator is the intended reading of the radius displayed on p. 10 of [2]. Therefore

$$\frac{2r_\eta}{\lambda} = O\left(\frac{\eta^{3/2}}{\log(1/\eta)}\right) = o(\eta).$$

Theorem 1 gives $m(S_\eta, \eta) = O(\eta^{-1})$.

This order is also necessary. The two axial endpoints have normalized geodesic distance

$$L_\eta = \frac{2}{\pi} \arctan\left(\frac{3q}{\lambda}\right) = \frac{2}{\pi} \arctan(6\pi\eta).$$

For all sufficiently small η ,

$$6\eta \leq L_\eta \leq 12\eta.$$

In an η -uniform tessellation, their Hamming distance is k/m for some integer $k \geq 1$ and satisfies

$$\frac{1}{m} \leq \frac{k}{m} \leq L_\eta + \eta \leq 13\eta.$$

We have proved:

Corollary 2 (deterministic/Gaussian separation). *For the family S_η in [2],*

$$m(S_\eta, \eta) = \Theta(\eta^{-1}).$$

At the same time $w_(S_\eta) \asymp 1$, while iid Gaussian hyperplanes need and suffice in number $\Theta(\eta^{-3}w_*(S_\eta)^2) = \Theta(\eta^{-3})$.*

Scope and full-upgrade audit

Route	Outcome
Latest Gaussian counterexample	Defeated deterministically by the axis-plane grid.
Thin tubes over low-dimensional bases	The sign-preserving retraction transfers any base tessellation with only twice the tube-radius loss.
Optimized direction measure	Promising reformulation as wedge quadrature, but no width-controlled equal-weight construction was found.
Deterministic lower bound from the Gaussian proof	Impossible by that mechanism: its random small-shift quotient is absent for axis-plane normals.
Entropy/communication lower bounds	Controlled at the conjectural scale by Sudakov; they do not produce a counterexample.
Multiscale chaining	Known net arguments retain a logarithm or an extra power of η ; sign discontinuity blocks the full upgrade.

Thus Corollary 2 resolves the strongest current candidate family in favor of deterministic hyperplanes, but it neither proves nor disproves the original conjecture for arbitrary K .

Novelty and verification. Exact-id, phrase, core-keyword, and later-citation searches through 11 August 2026 found the Gaussian counterexample but no statement of its deterministic $\Theta(\eta^{-1})$ tessellation. A numerical checker audits the projective interval count, the fiber-distance inequality, and the exact arc-length scaling. Human review should check the parameter unwinding from the final substitution in [2] and the source’s sign typo in the logarithmic radius.

References

- [1] Y. Plan and R. Vershynin, *Dimension reduction by random hyperplane tessellations*, arXiv:1111.4452. See Section 1.7.
- [2] S. Dirksen and N. Q. D. Strachan, *A resolution of the Gaussian hyperplane tessellation conjecture on the sphere*, arXiv:2508.05194. See the distinction on p. 2 and the lifted thin-cylinder construction on pp. 9–10.